



Gluco-con capsules

Description: Sz 0: caramel opaque/ caramel opaque.

Content:

Crystalline glucosamine sulphate sodium (equiv. to glucosamine sulphate 250mg) 314mg/cap.
Chondroitin sulphate 200mg/cap.

Indication:

As adjuvant therapy for osteoarthritis.

Pharmacology:

Glucosamine is an aminomonosaccharide, particularly effective in all forms of degenerative osteoarticular diseases e.g. arthrosis and osteoarthritis, both subacute and chronic, with disappearance or reduction of articular pains, improvement of articular pains, improvement of articular functions and inhibition or regression of the degenerative process. Glucosamine possesses arthrotrophic metabolic activity by stimulating the anabolic metabolism of osteocartilaginous tissues, through the stimulation of the biosynthesis of the mucopolysaccharides (which are essential components of the cartilage substances) and of the bone mesenchymal tissues, and improvement of the bone calcium uptake. Lubricant activity through improvement of synovial fluid viscosity by increasing synovial fluid production. Chondroitin sulphate belongs to a family of heteropolysaccharides called glycosaminoglycans (GAGs). It is a compound that appears to block the enzymes that can destroy crucial cartilage tissue. It is also believed to promote water retention and elasticity in joint cartilage. Its actions include promotion and maintenance of the structure and function of cartilage (referred to as chondroprotection), pain relief of osteoarthritic joints and anti-inflammatory activity.

After oral administration, only 26% of glucosamine is absorbed, with onset of action about 2-3 weeks for osteoarthritis. Bioavailability is low due to its first pass hepatic metabolism. Following oral administration, glucosamine is extensively metabolized in the liver. It is rapidly desulfated and metabolized into smaller molecules, and ultimately to carbon dioxide, water and urea. Glucosamine is not protein-bound, but rather incorporates into the plasma proteins (primarily globulin). Rapid diffusion into most tissues also occurs, including articular tissues. Urinary excretion accounts for 10% of an oral dose, about 11% is eliminated via the faeces, presumably as an unabsorbed drug. The elimination half-life of glucosamine which is incorporated into the plasma proteins is about 68 hours after oral doses. Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air in the lungs. Absorption of chondroitin appears to occur from the stomach and small intestines. Some chondroitin sulphate appears to enter the joint space.

Contraindications:

It is contraindicated in patients hypersensitive to glucosamine, chondroitin and allergic to shellfish.

Side effects/Adverse reactions:

Cardiovascular: Peripheral edema, tachycardia were reported in a few patients following a larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.



Central nervous system: Drowsiness, headache, insomnia has been observed rarely during therapy (less than 1%).

Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

Skin: Skin reactions such as erythema and pruritis have been reported with therapeutic administration of glucosamine.

Precautions:

Glucosamine sulphate is a causal therapy and the therapeutic effect is only seen after 1 week. Therefore, it is advisable to include an antiinflammatory drug (in case of severe pains) during the first days of treatment with glucosamine. Caution should be exercised to those with hemophilia and when taking with warfarin because of theoretical possibility that it may have antithrombotic activity. Children, pregnant women and nursing mothers should avoid using chondroitin sulphate.

Use in pregnancy and lactation:

It is contraindicated in children, pregnant ladies and nursing mothers.

Drug Interactions:

Oral hypoglycaemic agent (metformin, tolbutamide, rosiglitazone) - reduction in the effectiveness. Glucosamine is likely safe in patients with well-controlled diabetes (HbA 1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA 1c levels or those taking insulin, monitor blood glucose levels closely. Warfarin - theoretical possibility of increased bleeding when taking concomitantly. Chondroitin sulphate should not be used concomitantly with chitosan.

Dosage and administration:

Light to moderate symptoms - 2 capsules twice daily for at least 6 weeks. Severe symptoms - Initial, 2 capsules three times daily, of at least 8 weeks. Follow-up: 2 capsules twice daily for 3 - 4 months. Treatment should be repeated every other 6 months or less, according to medical prescription. Best taken before meals.

Treatment of overdose:

In the event of an acute overdose, the patient should be observed closely and symptomatic treatment should be instituted.

Derived from seafood.

Storage : Keep container tightly closed. Store in a dry place below 30°C.

Protect from light. Keep out of reach of children.

Presentation/Packing : 90 capsules, 60 capsules

Shelflife : 3 years

Product registration holder: PRIME PHARMACEUTICAL SDN. BHD.

& Manufacturer: 1505, Lorong Perusahaan Utama 1, Tmn. Perindustrian Bukit Tengah,
14000 Bukit Mertajam, Penang, Malaysia.

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